Python Project Report: Public Firm Analysis

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Introduction

This project examines the Business Services industry using both quantitative financial data and textual information from firms' 10-K filings. We selected Business Services because the sector includes a wide range of technology, software, payment, and professional service companies, making it well suited for comparing firm performance and analyzing changes in business priorities over time.

To define the industry sample, we used the major group classification provided in the dataset and selected firms with SIC codes beginning with 73, which corresponds to Business Services. The quantitative analysis focuses on firm characteristics, stock prices, sales, geographic distribution, return on assets, and historical financial performance. The text analysis examines firms' 10-K Item 1 disclosures using text cleaning, keyword frequency, TF-IDF, word clouds, and Word2Vec models.

We conduct the text analysis for both 2020 and 2025 to identify how the language and major themes in the Business Services sector changed over time. Finally, we perform a detailed analysis of one selected firm and use the results to develop a specific strategic recommendation. Together, these analyses provide a broader view of both the financial performance and evolving business environment of the Business Services industry.

Main findings

The sector contains 3,273 distinct firms and 30,610 firm-year observations. Data rises to the top of both sector keyword rankings in 2025, while platform gains prominence. For Paycom, revenue increases by 94.4% from 2021 to 2025 while its recorded stock price falls by 61.6%. We recommend testing an optional fixed-fee payroll automation package to capture a share of the customer value created by automation.

Data and analytical scope

The four course datasets provide annual financial records for 1994–2025, the SIC major-group lookup, and 10-K Item 1 text for 2020 and 2025 [1]. Steps A–B examine the financial panel; Steps C–F compare the two text samples; Step G examines Paycom's historical performance. All results below follow the final notebook and its exported tables. Industry rankings retain the source dataset's units; no cross-firm currency conversion or stock-split adjustment is applied. Paycom financial amounts are expressed in USD millions, as labelled in the notebook.

Step A Industry selection and data filtering

We selected major group 73 by retaining SIC codes from 7300 to 7399. The code normalizes gvkey to a common firm identifier, converts financial fields to numeric values, removes exact duplicate rows, and rejects conflicting records for the same firm and fiscal year. Counts use distinct gvkey–fyear pairs, preventing repeated records from inflating the sample.

Table 1. Business Services sample coverage

Measure	Result
Unique firm-year observations	30,610
Unique firms	3,273
Firms observed in every year from 1994 to 2025	64

Only 64 firms, about 2.0% of all sector firms, have records in all 32 years. The panel is therefore unbalanced: changes in annual averages can reflect entry, exit, and changes in sector membership.

Listing 1. Core sector filter and observation count

```
business_services = firms.loc[firms['sic'].between(7300, 7399)].copy()
firm_years = business_services[['gvkey', 'fyear']].drop_duplicates()
years_per_firm = firm_years.groupby('gvkey')['fyear'].nunique()
```

Step B Preliminary quantitative analysis

B1 Highest recorded stock prices

For each year, we exclude missing prices, sort prcc_c in descending order, and retain ten firms. These are per-share price rankings, which do not measure market capitalization, valuation multiples, or investment returns.

Table 2. Top ten stock prices in 2020 and 2025 in recorded price units

Rank	2020 firm	Price	2025 firm	Price
1	Alphabet	1,752.64	Constellation Software	2,399.74
2	Constellation Software	1,303.04	Fair Isaac	1,690.62
3	Shopify	1,131.95	United Rentals	809.32
4	CoStar Group	924.28	AppLovin	673.82
5	Trade Desk	801.00	Intuit	662.42
6	ServiceNow	550.43	Meta Platforms	660.09
7	Fair Isaac	511.04	Spotify Technology	580.71
8	Adobe	500.12	Mastercard	570.88
9	Paycom Software	452.25	CACI International Class A	532.81
10	Tyler Technologies	436.52	S&P Global	522.59

Alphabet ranks first in 2020; Constellation Software ranks first in 2025. Constellation and Fair Isaac appear in both top-ten lists. A change in these rankings can also reflect stock splits and sample composition.

B2 and B3 Sales leaders and geographic distribution

B2 Highest historical sales

To identify ten different firms, we sort all valid annual sales observations, retain each firm's peak-sales record, and then select the ten largest peaks. Thus, the result compares maximum annual sales across firms rather than the ten largest firm-year records or cumulative sales.

Table 3. Ten firms with the highest peak annual sales in dataset units

Rank	Firm	Peak year	Annual sales
1	Alphabet	2025	402,836.000
2	Microsoft	2025	281,724.000
3	Meta Platforms	2025	200,966.000
4	IBM	2011	106,916.000
5	Oracle	2025	67,357.000
6	Fujitsu	2010	54,717.315
7	NEC	1999	48,460.650
8	SAP	2025	43,207.817
9	Salesforce	2025	41,525.000
10	Lucent Technologies	1999	38,303.000

Alphabet, Microsoft, and Meta lead the ranking, with peaks in 2025. IBM's peak is in 2011, while NEC and Lucent peak in 1999. Historical peaks therefore capture different periods of scale and should not be read as a current competitive ranking.

B3 Top ten recorded locations

We count distinct gvkey values within each location. If a firm changed its recorded location over time, it is counted once in each location it occupied, so the location counts need not sum to the number of distinct firms.

Table 4. Geographic distribution by unique firms

Rank	Location	Code	Unique firms
1	United States	USA	2,684
2	China	CHN	152
3	Canada	CAN	80
4	Israel	ISR	76
5	United Kingdom	GBR	56
6	Hong Kong	HKG	39
7	Singapore	SGP	33
8	Japan	JPN	22
9	Ireland	IRL	15
10	Netherlands	NLD	11

The USA is the dominant recorded location, with 2,684 firms. China is second with 152, followed by Canada with 80. These records describe firm location; they do not show where firms earn revenue or operate their businesses.

B4 to B6 Historical prices crisis exposure and ROA

B4 uses the arithmetic mean of available stock prices in each fiscal year. B6 first selects location = USA, calculates ROA = net income / year-end assets for each record, and averages valid ratios by year. Zero assets and infinite ratios are excluded; extreme finite ratios are retained, matching the final code.

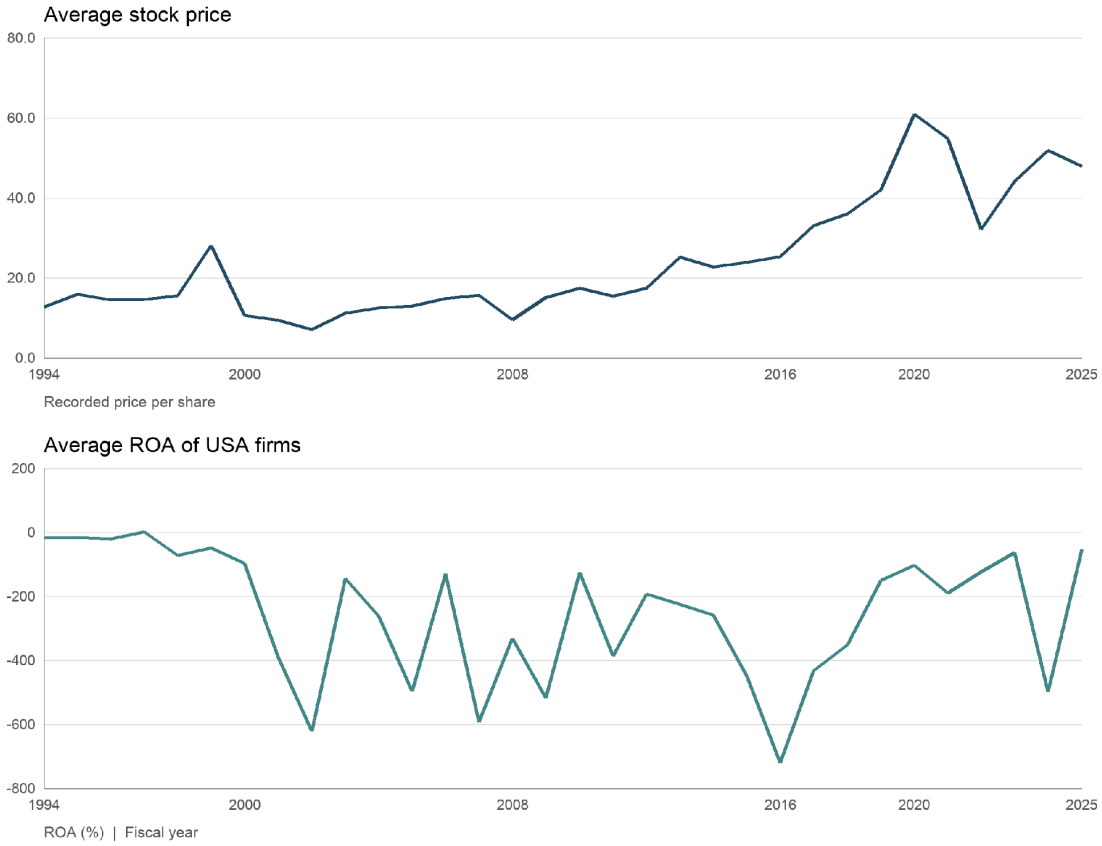


Figure 1. Annual mean stock price and USA mean ROA, 1994–2025. Source: public_firms.csv; charts redrawn from the notebook calculations.

The mean price falls from 15.59 in 2007 to 9.47 in 2008, reaches 60.94 in 2020, falls to 32.03 in 2022, and ends at 47.77 in 2025. This is a changing-sample price average, not a portfolio return index.

B5 Largest recorded decline in 2008

We match firms with valid prices in both 2007 and 2008 and require a positive 2007 price. Percentage drop = (price in 2007 − price in 2008) / price in 2007 × 100. The largest decline is for gvkey 183139, labelled Access Newswire Inc using its latest recorded company name: 1.25 to 0.0071, a 99.432% drop. The name may differ from its name in 2008. This identifies the largest observed decline; it does not isolate the crisis as its cause.

Interpreting the ROA series

The USA mean ROA is negative in 31 of the 32 years and reaches −721.76% in 2016. Individual ratios range from −3,495 to 683.2, making the arithmetic mean highly sensitive to extreme losses and small asset denominators. The chart should therefore be interpreted as the assignment's raw mean, not the profitability of a typical firm. A median or trimmed mean would be a useful robustness check, but is outside the reported code results.

Step C Text cleaning

C1 Clean the full filing sample

We load all 5,481 filing rows in the 2020 Item 1 dataset. Missing text is replaced with an empty string. A shared cleaning function converts text to lowercase, removes ASCII punctuation using string.punctuation, splits on whitespace, and removes NLTK English stop words [3]. The cleaned text is stored in a new item_1_clean column and the same function is reused for 2025.

Listing 2. Shared cleaning function

```
punctuation_table = str.maketrans('', '', string.punctuation)
def clean_text(text):
    text = str(text).lower().translate(punctuation_table)
    return ' '.join(w for w in text.split() if w not in stop_words)
```

This implements the required basic cleaning. It does not remove numbers or all Unicode symbols, split compounds formed when punctuation is deleted, or combine singular and plural forms. Consequently, company names, generic disclosure words, and artifacts such as •data can remain in the results. These limitations matter when interpreting keyword and similarity rankings.

Step D Keyword analysis for 2020

D1 One document per firm

We use gvkey as the merge key because it is the shared firm identifier; company names can change and are not reliable keys. Financial metadata is first restricted to fiscal year 2020, so joining to the full historical panel cannot multiply each filing across 32 years. Both inputs use normalized gvkey values.

Within each text year, missing identifiers and empty cleaned filings are excluded. When a firm has multiple filings, we retain the longest cleaned text; equal-length ties retain the original CSV order. Filing dates are unavailable, so this rule selects the most extensive text, not necessarily the latest filing. We remove 397 duplicate filing rows across the full 2020 sample; no missing identifiers or empty texts are excluded. An inner join followed by SIC filtering produces 482 sector firms.

Listing 3. Matching-year metadata and validated inner join

```
metadata = firms.loc[firms['fyear'].eq(year),
                     ['gvkey', 'fyear', 'conm', 'sic']]
merged = text.merge(metadata, on='gvkey', how='inner',
                    validate='one_to_one')
sector = merged.loc[merged['sic'].between(7300, 7399)].copy()
```

The one_to_one validation checks that each firm contributes exactly one document to the sector analysis. The inner join deliberately excludes firms without matching annual metadata; our text findings therefore describe the matched sample rather than every sector firm in the financial panel.

D2 and D3 Keyword rankings and word clouds

Word counts measure repetition within each firm's document. TF-IDF gives more weight to terms that distinguish one document from other sector documents. We use raw term frequency, smoothed inverse document frequency, and L2 normalization [4]. Each year's vocabulary and IDF are fitted separately. Rankings use nonzero terms and alphabetical tie-breaking; ten terms are exported for every firm.

Listing 4. Count and TF-IDF matrices from cleaned tokens

```
vectorizer = CountVectorizer(tokenizer=str.split,
                             token_pattern=None, lowercase=False)
counts = vectorizer.fit_transform(sector['item_1_clean'])
tfidf = TfidfTransformer().fit_transform(counts)
```

Table 5. Sector top ten terms in 2020 across 482 firms

Rank	Word-count term	Count	TF-IDF term	Sum
1	services	13,459	services	32.397
2	business	13,197	customers	29.110
3	customers	10,909	business	28.331
4	data	10,085	data	24.303
5	solutions	9,009	solutions	24.103
6	company	8,877	company	22.113
7	us	8,570	products	18.568
8	products	7,580	us	18.506
9	may	7,274	platform	17.838
10	including	7,126	customer	17.260

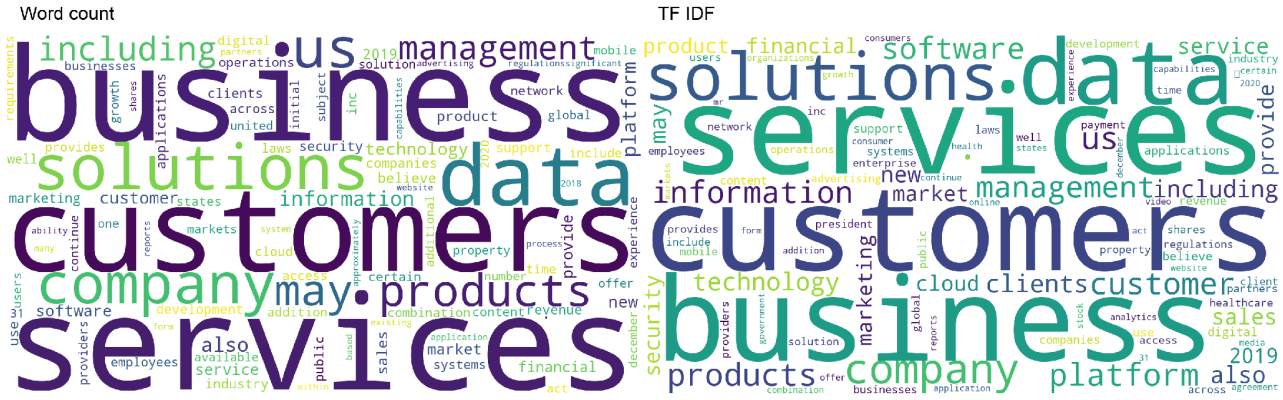


Figure 2. Business Services word clouds for 2020. Size reflects total word count or the sum of normalized firm TF-IDF scores; the largest 100 terms are displayed.

Services, customers, business, data, and solutions dominate both summaries. Platform appears in the TF-IDF top ten despite being only 14th by total count. This suggests that it is comparatively prominent within some firms' disclosures. Generic terms such as company and us remain because the cleaning uses the specified English stop-word list.

D4 Firm level keyword example

Comparing the two methods within a firm

The 2020 firm-level outputs contain 4,820 ranked rows per method: ten keywords for each of 482 firms. The following full ten-term example shows how the methods differ without requiring the reader to inspect the notebook. Scores are counts or normalized TF-IDF weights; they are not comparable on the same scale.

Table 6. Paysign's ten leading keywords in 2020 by each method

Rank	Word-count term	Count	TF-IDF term	Score
1	card	81	paysign	0.473
2	cards	64	card	0.369
3	laws	45	cards	0.360
4	prepaid	38	prepaid	0.267
5	paysign	37	debit	0.171
6	payment	35	plasma	0.149
7	information	31	pharmaceutical	0.130
8	regulations	31	payment	0.114
9	services	29	laws	0.106
10	payments	26	cardholder	0.104

Counts emphasize card, cards, and laws, while TF-IDF elevates Paysign's name and more distinctive terms such as debit, plasma, and pharmaceutical. Firm names can be discriminating statistically without being informative industry themes.

Step E Word embeddings for 2020

E1 Train Word2Vec on the full sample

We train the 2020 Word2Vec model on all 5,481 cleaned filing rows, including firms outside Business Services. Unlike the keyword stage, training retains duplicate filing rows, so repeated filings may receive extra influence. Each filing is treated as one token sequence. The model uses 100-dimensional vectors, a context window of five, a minimum token count of five, one worker, and seed 42. With no sg argument, Gensim uses its default CBOW architecture [5]. Step F repeats this procedure for 2025.

Listing 5. Full-sample embedding model and five-word query

```
sentences = [t.split() for t in cleaned[year]['item_1_clean']
             if t.strip()]
model = Word2Vec(sentences, vector_size=100, window=5,
                 min_count=5, workers=1, seed=42)
model.wv.most_similar('services', topn=5)
```

We select services, customers, and data because they are prominent in the sector clouds and represent service delivery, customer relationships, and information. Cosine similarity identifies words used in similar contexts; it does not imply that the words are interchangeable or that firms using them are competitors.

E2 Five nearest words for the 2020 keywords

We query services, customers, and data against the full-sample 2020 model. Table 7 lists all five neighbors for each keyword, with cosine similarities in parentheses. Higher scores indicate more similar usage contexts; they do not measure commercial relationships.

Table 7. Word2Vec neighbors for 2020 with cosine similarities

Rank	Services	Customers	Data
1	solutions (0.696)	clients (0.832)	information (0.598)
2	service (0.669)	customer (0.771)	datasets (0.576)
3	assistance (0.581)	endusers (0.751)	•data (0.574)
4	clients (0.529)	users (0.706)	evidence (0.565)
5	consulting (0.507)	consumers (0.702)	dataset (0.547)

Solutions and service are the closest neighbors to services, while clients and customer are closest to customers. Information and datasets are the closest neighbors to data. The token •data is a cleaning artifact, illustrating why even highly ranked model outputs require interpretation. Step F applies the same model settings to 2025 and compares the results.

Step F Comparative analysis for 2020 and 2025

F1 Repeat text cleaning and keyword analysis for 2025

We repeat the same cleaning, deduplication, same-year inner join, tokenization, keyword extraction, and embedding procedures for 2025. Of 4,536 full-sample filing rows, 27 duplicate filing rows are removed for firm-level keyword analysis; no missing identifiers or empty texts are excluded. The final sector sample contains 510 firms and 2,079,879 cleaned tokens, compared with 482 firms and 1,753,989 tokens in 2020.

Table 8. Sector top ten terms in 2025 across 510 firms

Rank	Word-count term	Count	TF-IDF term	Sum
1	data	14,039	data	30.587
2	services	12,972	services	29.994
3	customers	11,830	customers	29.967
4	business	11,565	business	25.927
5	company	10,156	solutions	24.091
6	platform	9,856	platform	22.676
7	us	9,618	us	21.203
8	solutions	9,304	company	20.952
9	including	8,727	including	19.090
10	also	7,790	information	18.292

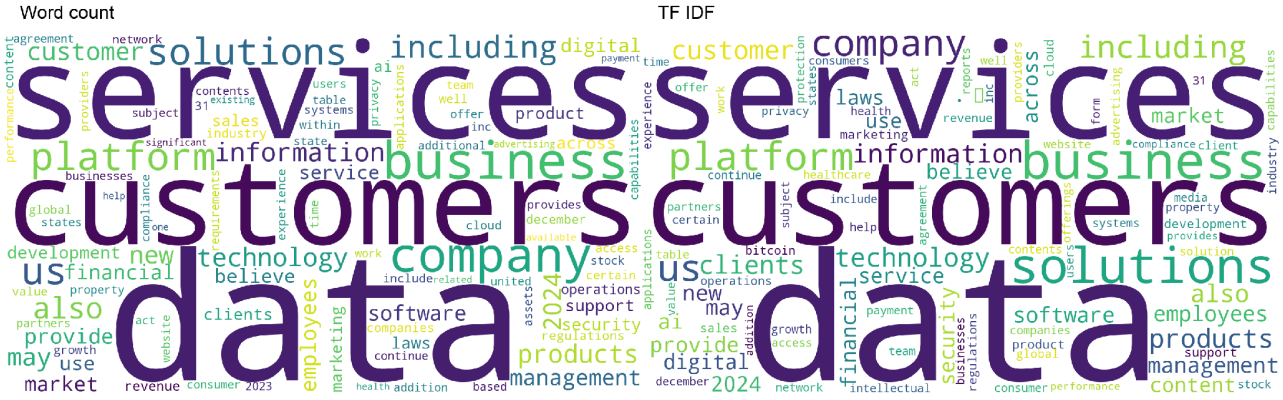


Figure 3. Business Services word clouds for 2025 using the same weighting definitions as Figure 2. Each cloud is scaled independently, so font sizes should not be compared numerically across years.

The main change in emphasis

Data moves from fourth to first under both word counts and aggregate TF-IDF. Its total count rises from 10,085 to 14,039. Platform rises from 14th to sixth by count and ninth to sixth by TF-IDF. Services remains highly prominent but falls from first to second. Business falls from second to fourth by count and from third to fourth by TF-IDF.

The clouds are generated from all terms across all matched sector firms, rather than from only their top-ten keyword lists. They give a concise overview of the industry language; exact ranks and normalized token shares provide stronger evidence for comparisons than cloud layout or font size alone.

F2 Word similarity comparison

We train the 2025 Word2Vec model on all 4,536 cleaned filing rows using the same settings as Step E. Table 9 reports its five neighbors for each keyword, with cosine similarities in parentheses. We compare these with the 2020 results in Table 7.

Table 9. Word2Vec neighbors for 2025 with cosine similarities

Rank	Services	Customers	Data
1	solutions (0.649)	clients (0.823)	information (0.605)
2	service (0.647)	customer (0.767)	datasets (0.580)
3	turnkey (0.530)	consumers (0.712)	dataset (0.577)
4	consulting (0.481)	endusers (0.710)	database (0.543)
5	ancillary (0.479)	users (0.694)	evidence (0.540)

Services

Solutions and service remain the two closest words. Assistance and clients leave the top five, while turnkey and ancillary enter in 2025. This is consistent with language describing packaged or supplementary services. It is an interpretation of disclosure context, not evidence that firms adopted a particular business model.

Customers

The same five neighbor words appear in both years: clients, customer, consumers, endusers, and users. Clients remains the closest term, with similarities of 0.832 and 0.823. The stable vocabulary suggests continuity in the way firms describe customer relationships, even though some neighbor ranks change.

Data

Information and datasets remain the first two neighbors. Database enters in 2025, while the artifact •data disappears from the top five. This is consistent with information-management language becoming more visible, but the artifact also demonstrates the limits of ASCII-only punctuation cleaning.

The models use different full samples, so neighbor ranks support descriptive comparisons rather than precise measurements of semantic change. Fixed seed and one worker reduce variation but do not guarantee identical outputs across software versions. The saved run contains Gensim runtime warnings; we report its exported neighbors with this limitation.

F3 Comparing industry themes with token shares

Raw totals rise partly because the 2025 sample has more firms and about 18.6% more cleaned tokens. To compare emphasis, we divide each keyword's count by all cleaned tokens in its year. This controls for total corpus size, although longer filings still carry more weight.

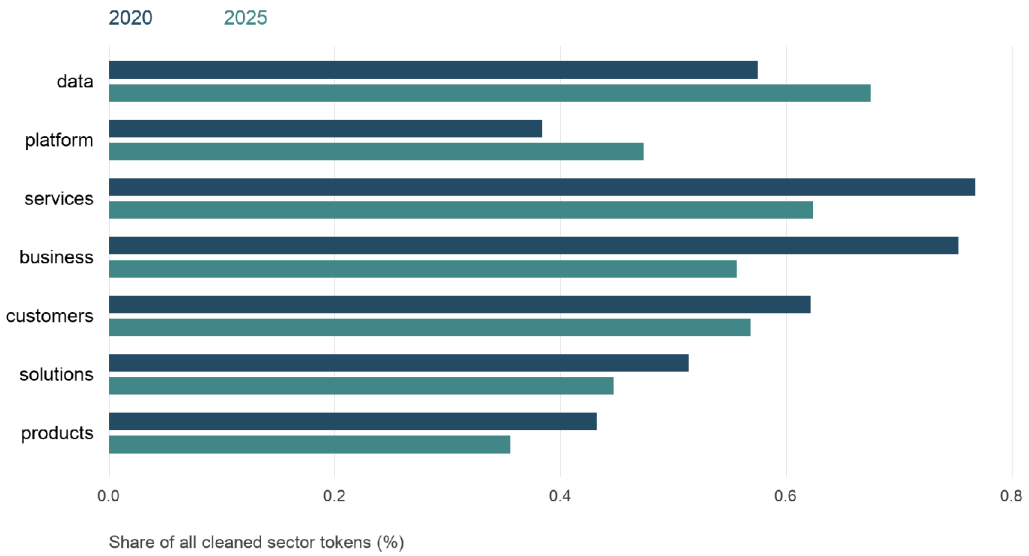


Figure 4. Selected keyword shares in the sector corpus. Source: F_keyword_comparison.csv. Shares measure token frequency, not the percentage of firms using a term.

Table 10. Changes in normalized keyword shares and word-count ranks

Keyword	2020 share	2025 share	Change in pp	Count rank
data	0.575%	0.675%	+0.100	4 → 1
platform	0.383%	0.474%	+0.090	14 → 6
services	0.767%	0.624%	-0.144	1 → 2
business	0.752%	0.556%	-0.196	2 → 4
customers	0.622%	0.569%	-0.053	3 → 3
solutions	0.514%	0.447%	-0.066	5 → 8
products	0.432%	0.356%	-0.076	8 → 13

Interpretation and boundaries

Data increases from 0.575% to 0.675% of all tokens, and platform increases from 0.383% to 0.474%. Their stronger normalized shares and higher ranks are consistent with increased attention to data and platform-based offerings. The decline in services and business indicates a relative change in language; it does not establish that firms provide fewer services.

Only 306 firms occur in both sector text samples. The comparison combines within-firm changes with entry, exit, changes in SIC classification, and availability of filings. TF-IDF is fitted separately each year, so we emphasize rankings rather than absolute score differences. Moreover, Item 1 describes businesses rather than measuring spending or realized outcomes. The results support a cautious conclusion that data and platform language became more prominent in the matched disclosures; they do not establish a causal industry transformation or specifically an AI investment trend.

Step G Paycom firm analysis and strategy

We select Paycom Software Inc, gvkey 20116 and SIC 7370, and focus on 2021–2025. The 2020 record is used only to calculate 2021 revenue growth. We inspect revenue, net income, assets, ROA, and year-end stock price to understand how operating growth and the recorded share price diverged. This is a historical firm analysis, not a peer-based market-share estimate.

Listing 6. Revenue growth and assignment-defined ROA

```
paycom = paycom.sort_values('fyear')
paycom['revenue_growth_pct'] = (
    paycom['sale'].pct_change(fill_method=None) * 100)
paycom['ROA_pct'] = (
    paycom['net_income'] / paycom['asset'].replace(0, np.nan) * 100)
```

Table 11. Paycom historical performance; monetary amounts in USD millions except per-share prices

Year	Revenue	Growth %	Net income	Assets	ROA %	Price $
2021	1,055.5	25.4	196.0	3,215.1	6.09	415.19
2022	1,375.2	30.3	281.4	3,902.5	7.21	310.31
2023	1,693.7	23.2	340.8	4,197.5	8.12	206.72
2024	1,883.2	11.2	502.0	5,859.9	8.57	204.97
2025	2,051.7	8.9	453.4	7,598.7	5.97	159.36



Figure 5. Paycom revenue, recorded stock price, assets, and ROA, 2021–2025. Source: G_Paycom_historical_performance.csv.

Revenue rises from $1,055.5 million to $2,051.7 million, a 94.4% increase, but annual growth slows from 30.3% in 2022 to 8.9% in 2025. Assets rise 136.3%, outpacing revenue growth. ROA peaks at 8.57% in 2024 and falls to 5.97% in 2025, while net income falls from $502.0 million to $453.4 million. The 2025 ROA decline reflects both lower net income and a larger asset base; the data alone do not identify the underlying operational cause.

G2 One strategy proposal for Paycom

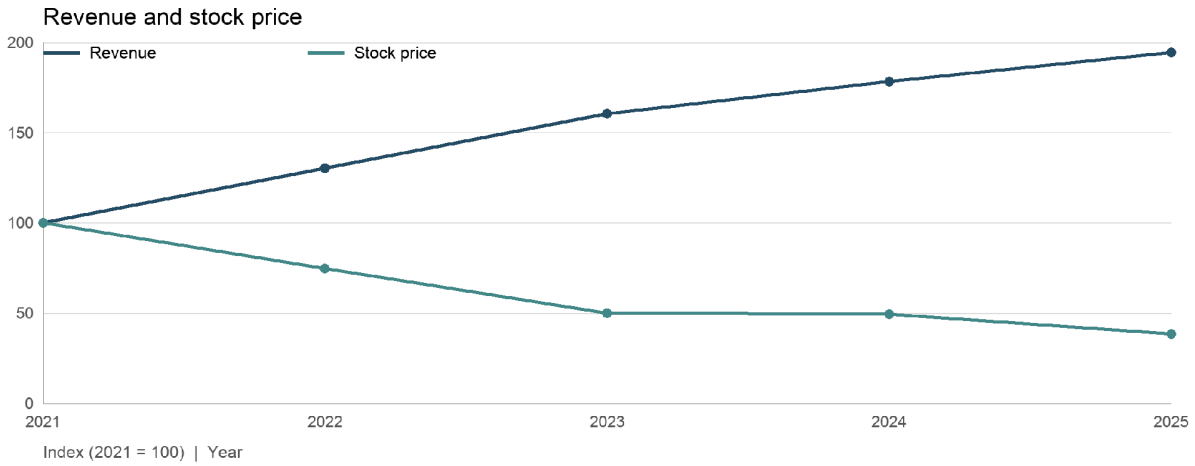


Figure 6. Paycom revenue and stock price on a common scale. Revenue ends at 194.4 and stock price at 38.4 when 2021 = 100.

The stock price falls 61.6% from $415.19 in 2021 to $159.36 in 2025 despite higher revenue. Across five annual observations, revenue and stock price have a Pearson correlation of −0.980; revenue and assets have a correlation of 0.906. These descriptive relationships are based on a very small time series and trending variables. They cannot explain the share-price decline or demonstrate that revenue growth caused it.

Business context from company disclosures

Paycom's 2023 annual report explains that Beti reduces payroll errors, billable corrections, and unscheduled payroll runs that would otherwise generate additional revenue. It also identifies lower sales of additional applications and lower retention among smaller clients as constraints on revenue growth, and describes a highly competitive HCM market [6, pp. 49, 54]. This supports a possible mechanism through which automation creates customer value while reducing some fee-generating activity. It does not quantify the contribution of these factors to the observed stock-price decline.

Recommendation Pilot an optional fixed fee automation package

Paycom should test an optional fixed-fee payroll automation package when customers review or renew their service arrangements. The package should price a defined set of automation benefits using evidence of fewer payroll errors and less administrative work. The goal is to capture customer value through predictable fees while avoiding dependence on correction-related activity. This is a proposed pilot, not an outcome demonstrated by the dataset.

Table 12. Proposed pilot measures and decision criteria

Measure	What the pilot should establish
Adoption and customer savings	Customers choose the package and achieve fewer corrections and fewer administrative hours.
Net revenue per customer	Package fees exceed lost correction-related revenue after incremental delivery costs.
Retention at service review	Retention remains acceptable relative to similar customers on existing pricing.

Compare pilot customers with a comparable group on existing pricing, matching customer size and prior usage where possible. Expand only if the package improves net economics while preserving customer value and retention. Customer-level usage, costs, and pricing records are needed to evaluate the proposal; the five-year financial dataset cannot test its effectiveness.

References and generative AI disclosure

References

[1] BAIT 508 course datasets. public_firms.csv; major_groups.csv; 2020_10K_item1_full.csv; 2025_10K_item1_full.csv. Course-provided data. Financial tables and text results are generated by BAIT508_Public_Firm_Analysis.ipynb.

[2] BAIT 508 Group Project: Public Firm Analysis, v4.0. Project instructions. Guidelines for Using Generative AI Tools in Group Projects, September 25, 2024.

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Generative AI use and evaluation

OpenAI Codex assisted with consolidating member code, writing English comments, removing duplicate operations, portability fixes, validation, and supplementary Step G, as disclosed in the notebook. For this report, Codex read the assignment and final notebook, drafted explanations and interpretations, formatted editable code excerpts and tables, redrew charts from the supplied data, and checked the rendered Word layout. The supplied cover details and introduction were retained, with "one" added before "selected firm" for grammar.

AI-generated explanations were checked against notebook outputs and exported CSVs. The report distinguishes observed results from proposed interpretations, identifies extreme ROA values and sample-composition limits, and checks the Beti revenue mechanism against Paycom's annual report. The strategy is a proposed test, not an empirically established solution. The group must review the final narrative, references, and code interpretation before submission.

Table 13. AI attribution record to complete before submission

Required field	Attribution
Tool and company	OpenAI Codex, OpenAI
Chat objective	Code integration and validation; report drafting, copyediting, figures, and Word formatting.
Use of AI-generated content	Integrated notebook support and report content described above.
Date and time	October 9, 2026, Vancouver
URL of chat history	https://chatgpt.com/s/cx_6ac993da58708191a8f040580a88ebb9 https://chatgpt.com/share/6ac993f8-0f90-83e8-aeb3-02e255535075